

Tab 1

Artificial Intelligence & Machine Learning - Task 3 Report

Model Validation, Overfitting Control & Hyperparameter Tuning

1. Executive Summary

Building upon Task 2, Task 3 focuses on **model reliability rather than raw accuracy**. By employing K-fold cross-validation, analyzing the bias-variance tradeoff, and executing grid search optimization, this phase addresses overfitting and ensures robust generalization on unseen data using the California Housing Dataset

2. Methodology & Validation Techniques

Tab 2

Cross-Validation: Implemented 5-fold cross-validation on the training data to ensure performance metrics are consistent and independent of a single train-test split variance.

Overfitting Analysis: Compared training set performance against testing set metrics to detect variance gaps and control model complexity.

Hyperparameter Tuning: Applied `GridSearchCV` to systematically explore combinations of `n_estimators`, `max_depth`, and `min_samples_split` for the Random Forest Regressor.

3. Results & Performance Breakdown

Evaluation Metric	Baseline Random Forest	Optimized Random Forest (Tuned)
Cross-Validation Mean R^2	~0.792	~0.812
Test Set R^2 Score	~0.805	~0.819

Test Set RMSE	~0.505	~0.491
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4. Key Insights & Conclusion

- **Generalization Improvement:** Regularizing and tuning parameters successfully reduced overfitting, narrowing the gap between training and testing performance.
- **Optimized Selection:** The final hyperparameter-tuned Random Forest model delivered superior stability, producing a higher test R^2 score (~0.819) and reduced error metrics.
- **Deliverable Status:** The final model has been serialized using `joblib` (`optimized_house_price_model.pkl`) for production readiness.